

TOA-Assignment 2

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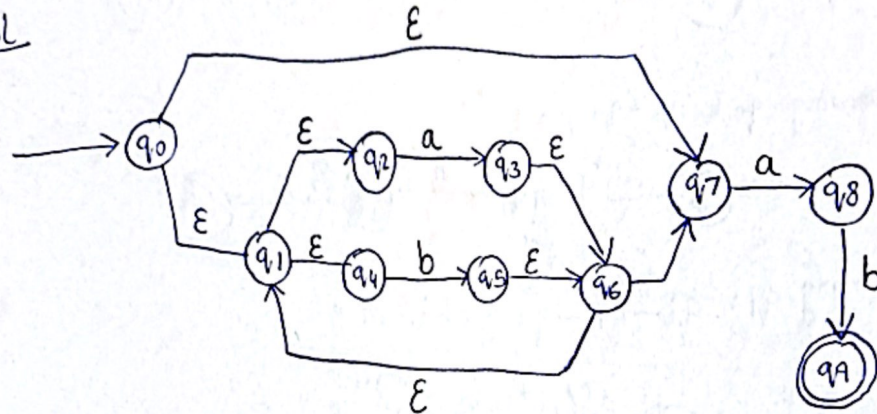
Class :- BCS-3E

Section A :-

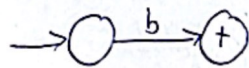
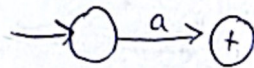
pg 1

Q1a) Regular Expression :- $(a+b)^*ab$

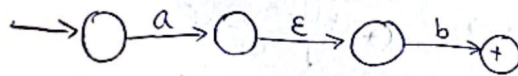
Sol



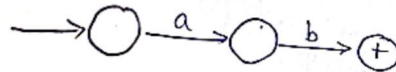
working of sub diagrams



so ab (concatenate)

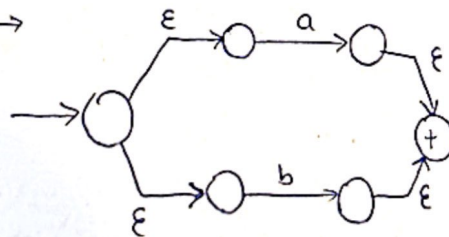


simplify

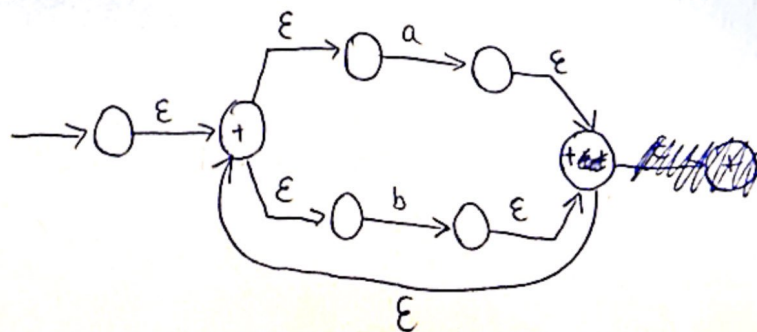


for $(a+b)^*$

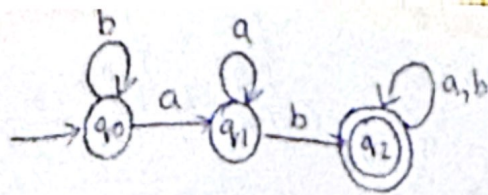
$a+b \rightarrow$



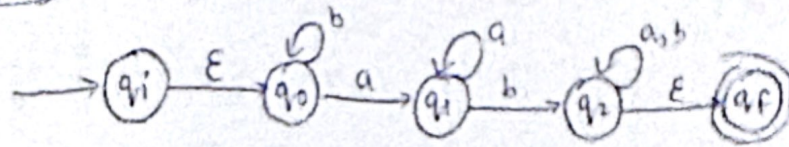
now $(a+b)^*$



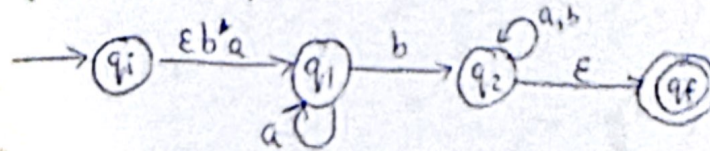
Q1b)



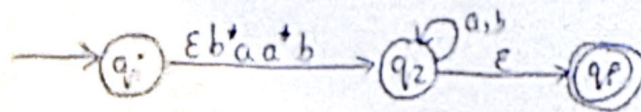
working



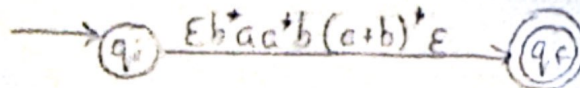
Removing q_0 : $q_i - q_0 q_1 = \epsilon b^* a$



Removing q_1 : $q_i - q_1 - q_2 = \epsilon b^* a \cdot a^* b$

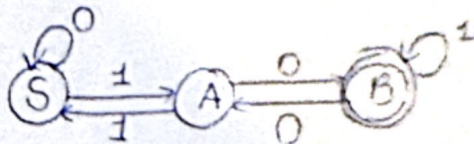


Removing q_2 : $q_i - q_2 - q_f = (\epsilon b^* a a^* b) \cdot (a+b)^* \epsilon$

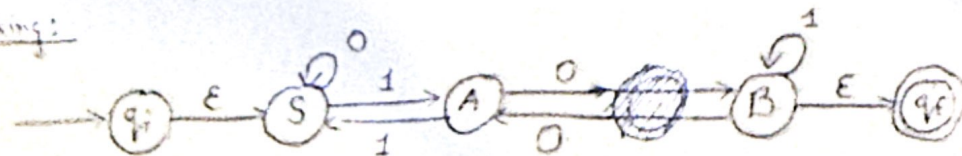


$\therefore$ Regular Expression : $b^* a a^* b (a+b)^*$ Ans

Q2a)



working:



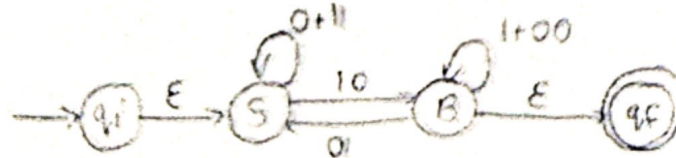
Removing A:

SAB : 10

BAS : 01

SAS : 11

BAB : 00

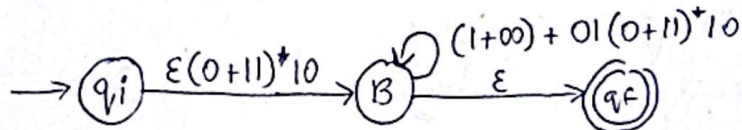


Removing S :

pg 3

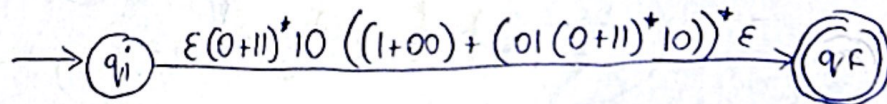
$$q_i - S - B = \epsilon (0+1)^* 10$$

$$B - S - B = (1+00) + 01(0+1)^* 10$$



Removing B :

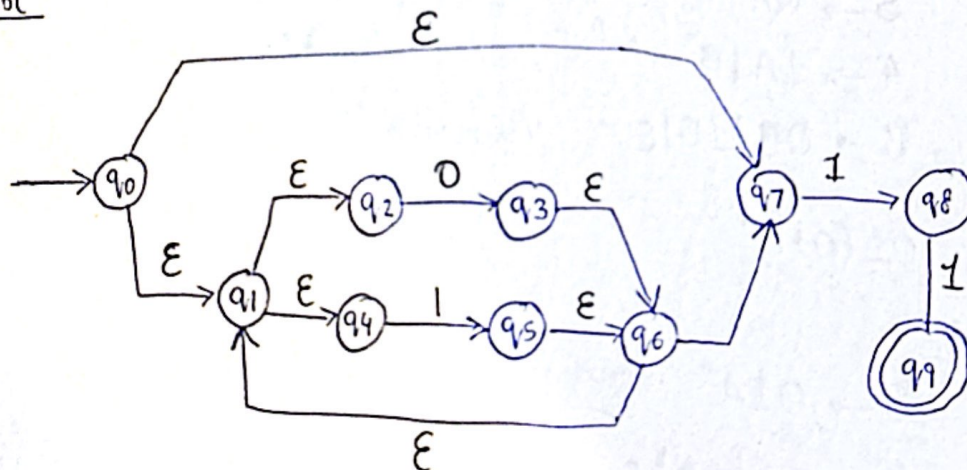
$$q_i - B - q_f = \epsilon (0+1)^* 10 ((1+00) + (01(0+1)^* 10))^* \epsilon$$



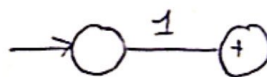
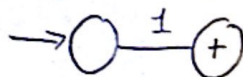
$\therefore$ Regular Expression : $(0+1)^* 10 ((1+00) + (01(0+1)^* 10))^* \epsilon$ Ans

Q2b) Regular Expression :- $(0+1)^* 11$

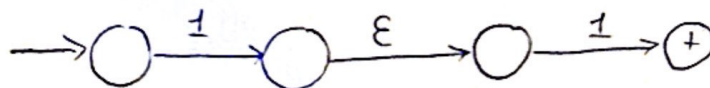
Sol



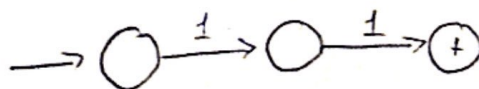
working of sub diagram



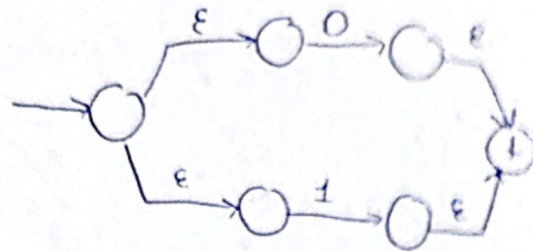
so 11 concatenation



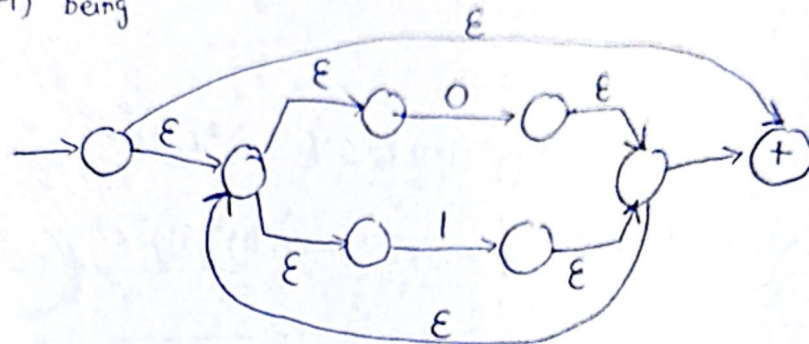
simplify



for $(0+1)^*$



$\therefore (0+1)^*$ being



Section B:

Q1i) $01^*[0+1]^*$

sol

$$S \rightarrow 0A$$

// will begin with 0 always

$$A \rightarrow 1A \mid B$$

// will generate 1^* or jump to B

$$B \rightarrow 0B \mid 1B \mid \epsilon$$

// Generates $(0+1)^*$ ends in ϵ

ii) $01(0+1)^*$

sol

$$S \rightarrow 01A$$

$$A \rightarrow 0A \mid 1A \mid \epsilon$$

iii) Letters followed by digits

$$L = \{xy \mid x \in \{a, \dots, z, A, \dots, Z\} \mid y \in \{0-9\}^*\}$$

my assumption string is of two length, and small + capital letter

sol

$$S \rightarrow aQ_0 \mid bQ_0 \mid \dots \mid zQ_0 \mid AQ_0 \mid BQ_0 \mid \dots \mid ZQ_0$$

$$Q_0 \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$$

i) Start and end with a.

pg 5

Regular expression: $a(a+b)^*a$

assuming Alphabet $\{a, b\}$

Sol

$$S \rightarrow aA | a$$

// always start with a, will accept single a

$$A \rightarrow aA | bA | a$$

// $(a+b)^*$ displays multiple time will always terminate at a

accepted string = (a, aaa, aababa, ...)

v) Generate all even integer upto 998

eg strings (0, 2, 4, ..., 10, 12, 14, ..., 102, 104, ..., 312, 314, ..., 996, ..., 998)

Sol

$$S \rightarrow 0 | X | NE | NDE$$

$$X \rightarrow 2 | 4 | 6 | 8$$

$$N \rightarrow 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9$$

$$E \rightarrow 0 | 2 | 4 | 6 | 8$$

$$D \rightarrow 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9$$

where

X is single digit number being printed

NE is double digit number (10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98)

NDE is triple digit number (106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, 498, 506, 508, 510, 512, 514, 516, 518, 520, 522, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, 548, 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, 680, 682, 684, 686, 688, 690, 692, 694, 696, 698, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, 754, 756, 758, 760, 762, 764, 766, 768, 770, 772, 774, 776, 778, 780, 782, 784, 786, 788, 790, 792, 794, 796, 798, 806, 808, 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, 846, 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876, 878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 906, 908, 910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940, 942, 944, 946, 948, 950, 952, 954, 956, 958, 960, 962, 964, 966, 968, 970, 972, 974, 976, 978, 980, 982, 984, 986, 988, 996, 998)

vi) Alternate Letters

~~Assumpti~~ Assuming alphabet $\{a, b\}$

so Regular Expression: $(ab)^* + (ba)^*$

eg strings (ab, ababab, ba, baba, ...)

// will give $(ba)^*$

Sol

$$S \rightarrow A | B$$

$$A \rightarrow abA | \epsilon$$

// will give $(ab)^*$

$$B \rightarrow baB | \epsilon$$

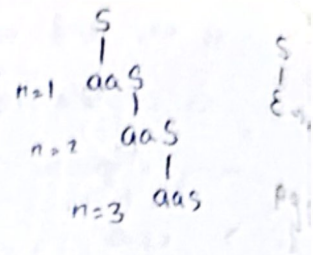
vii) a^{2n}

$L = \{a^{2n} \mid n \geq 0\}$

so Regular expression = $(aa)^*$

sol

$S \rightarrow aaS \mid \epsilon$



viii) $[ab]^n$

$L = \{(ab)^n \mid n \geq 0\}$

so Regular expression = $(ab)^*$

sol

$S \rightarrow abS \mid \epsilon$

ix) Generate $\{012, 120, 201\}$

sol 1

$S \rightarrow 012 \mid 120 \mid 201$

sol 2

$S \rightarrow 0A \mid 1B \mid 2C$

$A \rightarrow 1D$

$D \rightarrow 2$

$B \rightarrow 2E$

$E \rightarrow 0$

$C \rightarrow 0F$

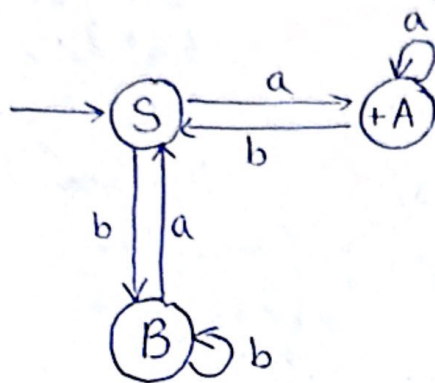
$F \rightarrow 1$

x) Grammar for the set of all strings over $\{0, 1\}$ consisting of equal number of 0's and 1's

$S \rightarrow 0S1 \mid 1S0 \mid 10S \mid 01S \mid \epsilon$

Q2a) $S \rightarrow aA \mid bB$
 $A \rightarrow aA \mid bS \mid \epsilon$
 $B \rightarrow bB \mid aS$

FA:



working:

Initial State : S

Non-terminal : {S, A, B}

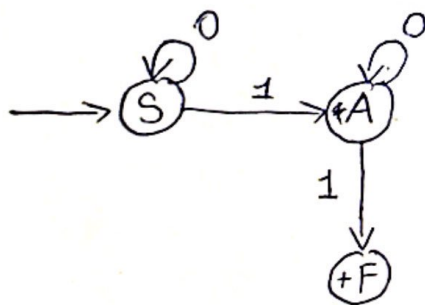
Terminal : {a, b}

Final : A as ϵ terminator

δ	a	b
S	A	B
A	A	S
B	S	B

b) $S \rightarrow OS \mid 1A$
 $A \rightarrow OA \mid 1$

sol



working:

Initial State : S

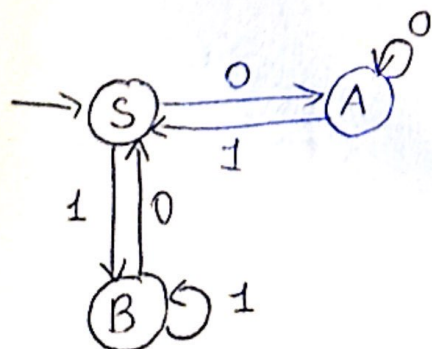
Final State : A as 1 terminate

Non-terminal : {A, S, F}

Alphabet/Terminal : {0, 1}

δ	0	1
S	S	A
A	A	F
F		

c) $S \rightarrow OA \mid 1B$
 $A \rightarrow 1S \mid OA$
 $B \rightarrow OS \mid 1B$



working:

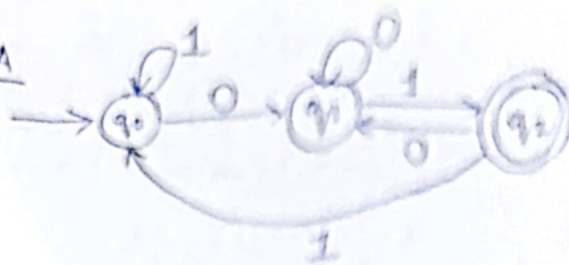
Initial State : S

Non-terminal State : {S, A, B}

Alphabet : {0, 1}

Final state :- none as no terminal or epsilon end, infinite loop

Q3a) DFA



Start State : Q_0
 Final State : Q_2
 Alphabet = $\{0, 1\}$
 Non-terminal = $\{Q_0, Q_1, Q_2\}$

Right-Linear Grammar [RLG]

$$Q_0 \rightarrow 0Q_1 \mid 1Q_0$$

$$Q_1 \rightarrow 0Q_1 \mid 1Q_2$$

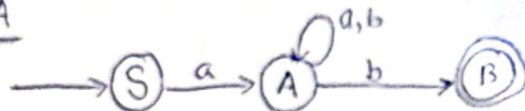
$$Q_2 \rightarrow 0Q_1 \mid 1Q_0 \mid \epsilon$$

CFG

$$S \rightarrow X01$$

$$X \rightarrow 0X \mid 1X \mid \epsilon$$

b) NFA



Start State : S
 Final State : B
 Alphabet = $\{a, b\}$
 Non-terminal = $\{S, A, B\}$

Right-Linear Grammar [RLG]

$$S \rightarrow aA$$

$$A \rightarrow aA \mid bA \mid bB$$

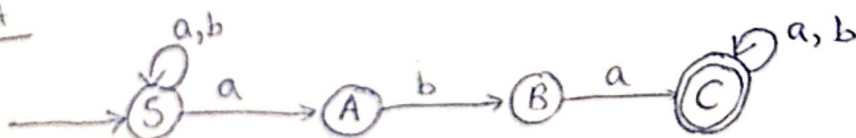
$$B \rightarrow \epsilon$$

CFG

$$S \rightarrow aMb$$

$$M \rightarrow aM \mid bM \mid \epsilon$$

c) NFA



Start State : S

Final State : C

Alphabet : $\{a, b\}$

Non-Terminal = $\{S, A, B, C\}$

Right Linear Grammar (RLG)

$$S \rightarrow aS | bS | aA$$

$$A \rightarrow bB$$

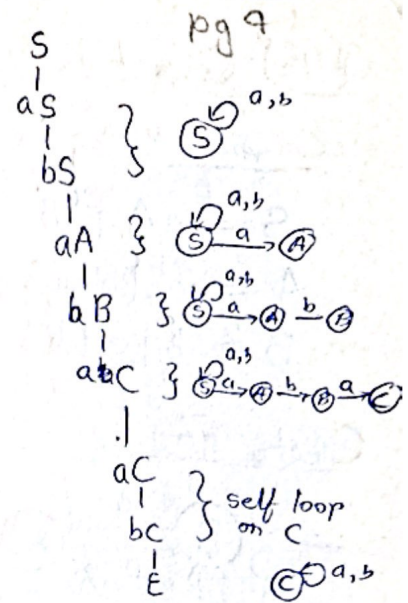
$$B \rightarrow \cancel{b} aC$$

$$C \rightarrow aC | bC | \epsilon$$

CFG

$$S \rightarrow XabaX$$

$$X \rightarrow aX | bX | \epsilon$$



Q4a) Converting LLG to RLG

Original LLG

$$S \rightarrow Aa | Ab$$

$$A \rightarrow Ba | Bb | a$$

$$B \rightarrow b$$

Find string produced by LLG

$$S \rightarrow Aa | Ab$$

$$A \rightarrow ba | bba$$

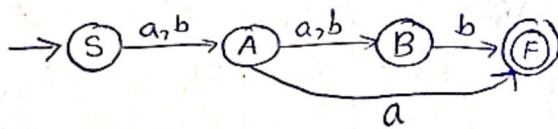
so

$$S \rightarrow baa | bba | aa | bab | bbb | ab$$

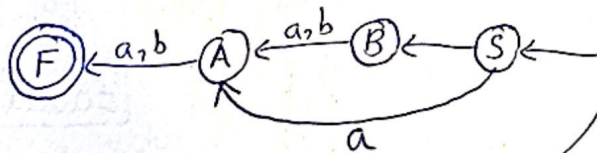
$$L = \{aa, baa, bba, ab, bab, bbb\}$$

generated

Create FA



Reverse FA



RLG:

$$S \rightarrow aA | bB$$

$$A \rightarrow aF | bF$$

$$B \rightarrow aA | bA$$

$$F \rightarrow \epsilon$$

S	S	S	S	S	S
aA	aA	bB	bB	bB	bB
aF	bF	aA	aA	bA	bA
ε	ε	aF	aF	bF	bF
ε	ε	ε	ε	ε	ε
= aa	= ab	= baa	= bba	= bab	= bbb

∴ The conversion was successful as strings made by LLG matches with strings made by RLG. You can see parse tree

Q4b) Convert the following RLG $\Rightarrow$ LLG

Pg 10

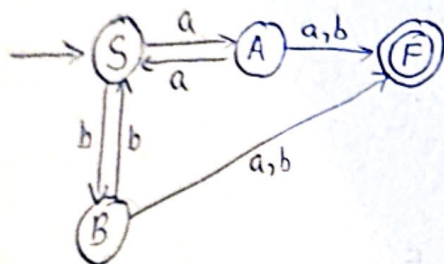
Original RLG

$S \rightarrow aA \mid bB$

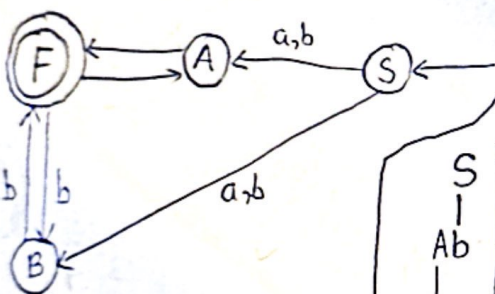
$A \rightarrow a \mid b \mid aS$

$B \rightarrow b \mid a \mid bS$

Create FA



Reverse FA



LLG:

$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow Fa$

$B \rightarrow Fb$

$F \rightarrow Aa \mid Bb \mid \epsilon$

if we simplify it then

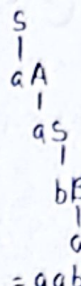
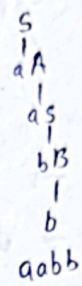
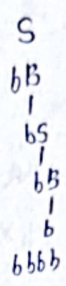
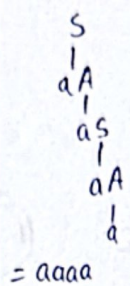
$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow a \mid Aaa \mid Bba$

$B \rightarrow b \mid Aob \mid Bbb$

Find some strings made by RLG to confirm with LLG string

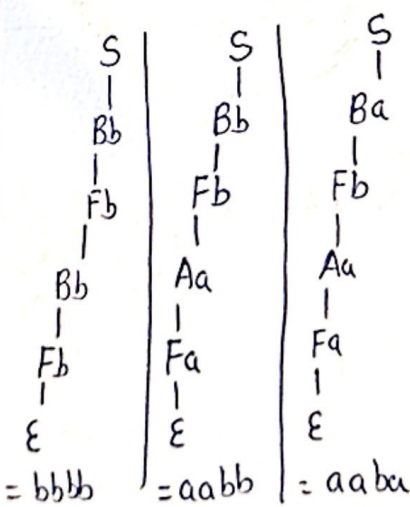
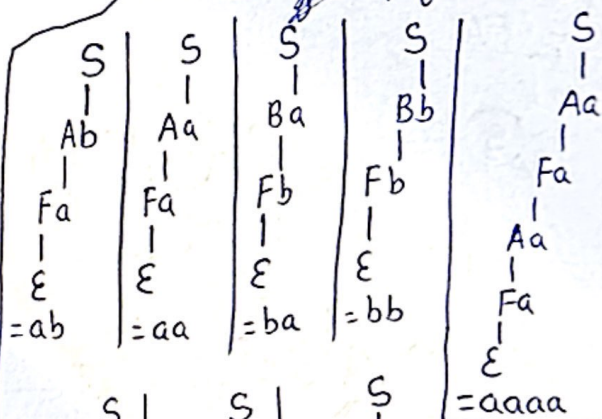
$\{aa, ab, ba, bb, aaaa, bbbb, aabb, aaba\}$



$\therefore$ Test case string

$aa, ab, ba, bb, aaaa, bbbb, aabb, aaba$

To check if LLG was correctly converted, let's confirm using parse tree. (using simplified LLG)



$\therefore$ The conversion was successful as strings made by RLG match with string made by LLG. You can see parse tree.

Q5) For 1

Pg 11

$S \rightarrow aSb \mid ab$

string = aabb

a) substitution

$S \rightarrow aSb$
 $a(ab)b$
 $aabb$

string achieved

b) Left most derivation

$S \rightarrow aSb$
 $a(ab)b$
 $aabb$

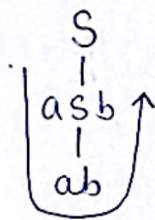
string achieved

Right most derivation

$S \rightarrow aSb$
 $a(ab)b$
 $aabb$

string achieved

c) Parse tree



= aabb

string achieved

Note = All three derivation coincide cause at each stage there is just one non-termining

Q5) For 2

$S \rightarrow aS \mid Sb \mid ab$

string = aaabb

a) substitution

$S \rightarrow aS$
 $a(aS)$
 $aa(Sb)$
 $aa(ab)b$
 $aaabb$

string achieved

b) Left most derivation

$S \rightarrow Sb$
 $(aS)b$
 $a(aS)b$
 $aa(ab)b$
 $aaabb$

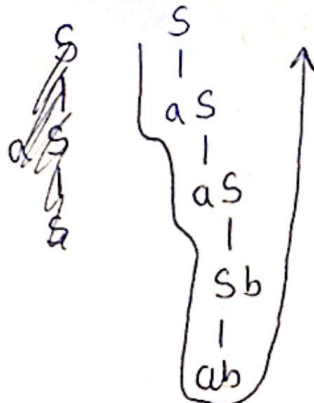
String achieved

Right most derivation

$S \rightarrow aS$
 $a(aS)$
 $aa(Sb)$
 $aa(ab)b$
 $aaabb$

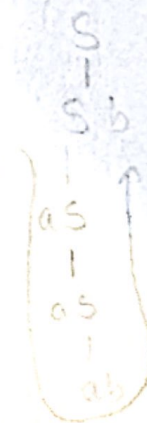
String achieved

c) Parse tree



= aaabb

String achieved



aaabb

Q5) For 3

pg 12

$S \rightarrow OS1|01$ string 000111

a) Substitution

$S \rightarrow OS1$

$O(OS1)1$

$OO(01)11$

000111

string achieved

b) Left most derivation

$S \rightarrow OS1$

$O(OS1)1$

$OO(01)11$

000111

string achieved

Right most derivation

$S \rightarrow OS1$

$O(OS1)1$

$OO(01)11$

000111

string achieved

c) Parse Tree



= 000111

string achieved

Q6i) CGF for balanced parenthesis in this alphabet set $\{(,)\}$

sol

CGF $\Rightarrow S \rightarrow (S) | SS | \epsilon$

eg string: $()$, $(())$...

ii a) CGF of even palindrome 001100

sol

CGF $\Rightarrow S \rightarrow OSO | 1S1 | \epsilon$

eg string: 001100, 010010, 011110, ...

ii b) CGF of odd palindrome 00100

sol

CGF $\Rightarrow S \rightarrow OSO | 1S1 | 0 | 1$

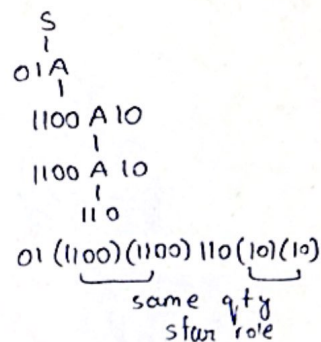
eg string: 01010, 01110, 00100, ...

$$1) \{01(1100)^n 110(10)^n \mid n \geq 0\}$$

sol

$$S \rightarrow 01A$$

$$A \rightarrow 1100A10 \mid 110$$



$$1) \{0^n 1^m \mid n = m\}$$

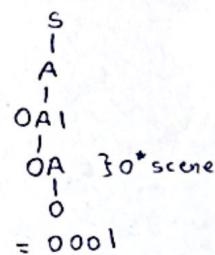
sol

so $n > m$ and $n < m$ means 0 more than 1 and 1 more than 0.

$$S \rightarrow A \mid B$$

$$A \rightarrow 0A1 \mid 0A \mid 0$$

$$B \rightarrow 0B1 \mid B1 \mid 1$$



g string ; 0, 1, 001, 011, 0001, 0111,

$$1) \{[0^n 1^n 2^m 3^m \mid n \geq 1, m \geq 1] \text{ or } [0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1]\}$$

sol

Language 1

Language 2

union of it in $S \rightarrow L_1 \mid L_2$

for Language 1

$$A \rightarrow XY$$

$$X \rightarrow 0X1 \mid 01$$

$$Y \rightarrow 2Y3 \mid 23$$

// generates $0^n 1^n \mid n \geq 1$

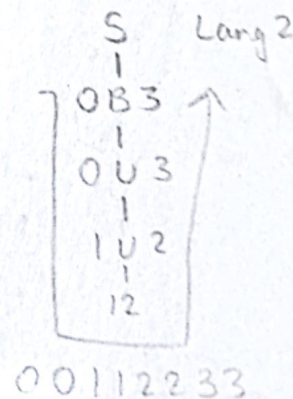
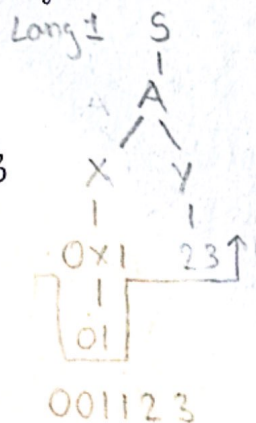
// generates $2^m 3^m \mid m \geq 1$

for Language 2

$$B \rightarrow \cancel{0B1} \mid 0U3$$

$$\cancel{01} \mid \cancel{02} \mid \cancel{03} \mid \cancel{04} \mid \cancel{05} \mid \cancel{06} \mid \cancel{07} \mid \cancel{08} \mid \cancel{09}$$

$$U \rightarrow 1U2 \mid 12$$



CFG

$$S \rightarrow A \mid B$$

// This will call language 1 grammar or language 2 grammar

vii) $\{ [0^i 1^j 2^k \mid k \leq i, k \leq j] \}$

$S_1 \rightarrow pa$

$p \rightarrow OP \mid \epsilon$

$a \rightarrow OR \mid R$

$r \rightarrow IR \mid \epsilon$

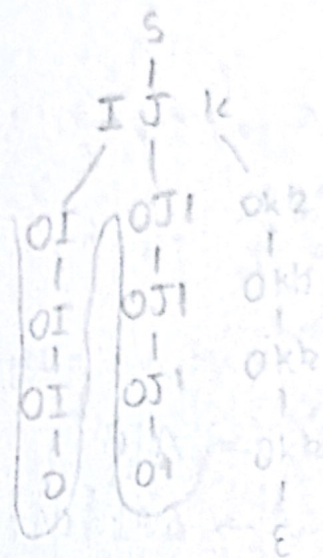
0 greater

CGF for $L_i(i > j > k)$

$S \rightarrow OS \mid T$

$T \rightarrow OT \mid U$

$U \rightarrow OU \mid OI$

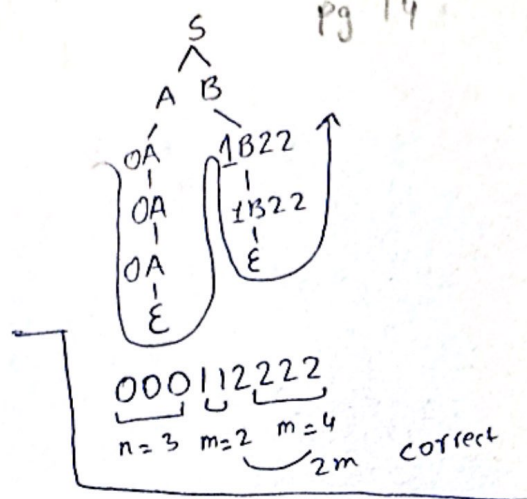


Part vi, vii, x are not context free so they are not solvable. Therefore skipping them

iii) $\{0^n 1^m 2^{2m} \mid n, m \geq 0\}$

sol

$$\begin{aligned} \text{CFG} \Rightarrow S &\rightarrow AB \\ A &\rightarrow 0A \mid \epsilon \\ B &\rightarrow 1B22 \mid \epsilon \end{aligned}$$

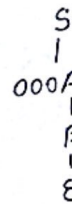


x) $\{0^n 1^m \mid n \leq m+3\}$

$$S \rightarrow 000A \mid 00A \mid 0A \mid A$$

$$A \rightarrow 0A1 \mid B \quad A \rightarrow 0A1 \mid B$$

$$B \rightarrow 1B \mid \epsilon \quad B \rightarrow 1B \mid \epsilon$$



$$000 \Rightarrow n=3$$

$$\text{so } n \leq 0+3 \text{ true}$$

Me) Input alphabet = $\{[,], \{, \}, (,)\}$

$$\text{CFG} \Rightarrow S \rightarrow (S) \mid [S] \mid \{S\} \mid SS \mid \epsilon$$

Section C

$$\begin{aligned} \text{CFG} \Rightarrow S &\rightarrow aSa \mid bSb \mid a \mid b \\ A &\rightarrow B \mid a \\ B &\rightarrow C \mid b \end{aligned}$$

1) Removing useless production from CFG

non-terminal states A, B, C are not reachable from S so removing them

$$\therefore S \rightarrow aSa \mid bSb \mid a \mid b$$

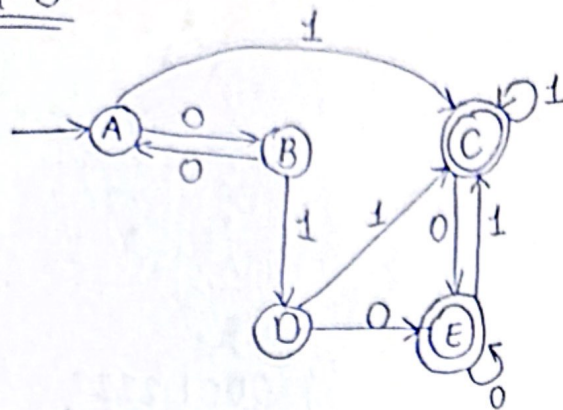
2) Check for ambiguity in the given CFG for the string 'abba'.

The CFG generates odd length palindrome, while string "abba" is of even length. Therefore, it ain't in the language.

As this string can't be generated, the question of ambiguity

Section D

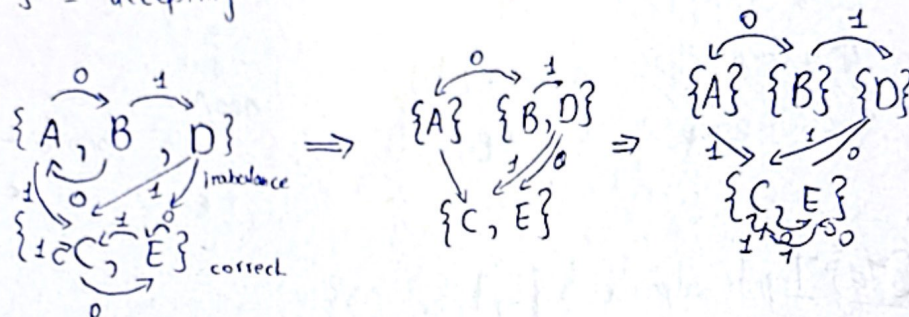
Q1)



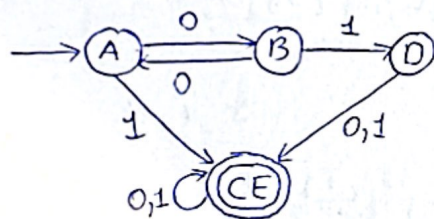
$\{A, B, D\}$ = Non-accepting

$\{C, E\}$ = accepting

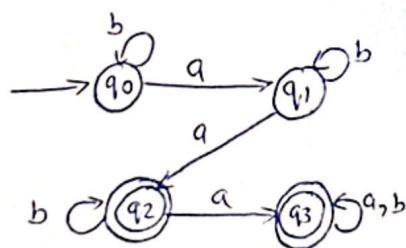
Sol



Minimised DFA

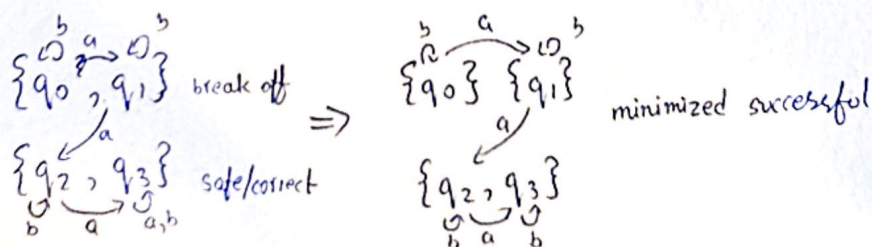


Q2) DFA



Accepting = $\{q_2, q_3\}$

Non-accepting = $\{q_0, q_1\}$



Minimised DFA

